Independent S_8 Compressed Proxies and the Optics Channel in the G1DM SRO Audit

Addendum to v0.2.1 Technical Note (v0.3)

G1DM Data Notes Toolkit v0.3

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Release relation: G1DM data-note supplement included with FDS–G1 v1.3.

1 Context

The v0.2.1 technical note established a source–response separation from public cosmological chains: Planck 2018 requires a nonzero CDM-like carrier floor ($T^D_{\mu\nu} \neq 0$), while DESI DR1 full-shape/RSD `_mu_sigma` chains do not require leading modified-growth leakage ($\mu_{\rm grav} \simeq 1$). The optics/Weyl channel remained scenario-dependent: a Planck-linked Weyl residual appears in FS/BAO+Planck (3.86σ) but is removed when DES Y3 3×2pt data are added.

2 v0.3: Independent S_8 compressed-proxy audit

Version 0.3 extends the SRO sparse audit (Note 4, Phase 2) by adding an independent S_8 tension proxy from public weak-lensing data. $S_8 = \sigma_8(\Omega_m/0.3)^{0.5}$.

Table 1: Independent S_8 compressed proxies.

Source	S_8	σ	$z(\Delta S_8)$
Planck 2018 (TT,TE,EE+lowE)	0.831	0.013	—
KiDS-1000 cosmic shear (fiducial)	0.759	0.0225	2.77σ
DES Y3 3×2pt	0.776	0.017	2.57σ

The S_8 observable row is loaded onto the response and optics channels as $[0, r, 1-r]$ with a response-loading fraction $r \in \{0, 0.25, 0.5, 0.75, 1.0\}$. At $r = 0$, the full S_8 tension is assigned to optics; at $r = 1$, to response; intermediate values represent mixed attribution. The source row ($\Omega_c h^2$, hard floor) and response row (μ_0 from DESI+DESY3joint, $z = 0.17$) are unchanged from v0.2.2.

†: At $r = 1$, the response channel absorbs the full S_8 tension. This reflects artificial response pressure from the proxy assignment, not a growth-leakage detection (the DESI μ_0 row remains near zero).

Source+optics is consistently preferred across the full r-grid for both KiDS and DES Y3 proxies. All results are verified with the source z-score capped at 10, confirming rankings are stable and not dominated by the 100σ carrier-floor row.

Table 2: SRO model comparison with S_8 proxy (r-grid summary).

Proxy	r	Best model (BIC)	ΔBIC(opt vs src)	ΔBIC(resp vs src)
KiDS-1000	0.00	source+optics	−6.57	+0.40
KiDS-1000	0.50	source+optics	−6.57	−2.13
KiDS-1000	1.00	source+optics†	—	—
DES Y3	0.00	source+optics	−5.51	+0.48
DES Y3	0.50	source+optics	−5.51	−1.72
DES Y3	1.00	source+optics†	—	—

3 Interpretation

The independent S_8 compressed proxies select an optics/low-redshift lensing-structure component beyond the mandatory source floor. This result is r-stable: it does not depend on an arbitrary loading-fraction choice. The DESI μ_0 row and the caps-around-$r = 1$ behavior confirm that the growth/response channel is not required.

The combined v0.2–v0.3 compressed diagnostic is therefore:

$$\boxed{T^D_{\mu\nu} \neq 0, \qquad \mu_{\rm grav} \simeq 1, \qquad \mathcal{D}^{S_8}_{\rm optics} \neq 0 \text{ at compressed-proxy level.}} \tag{1}$$

Caveats:

- This is a compressed Gaussian SRO audit, not a production multi-probe likelihood. Full 3×2pt covariance, pipeline cross-correlation, and systematic nuisance profiling are required for a production-level result.
- The S_8 tension proxy is a known low-redshift structure tension diagnostic; it does not directly measure a Weyl/lensing residual in the G1DM sense. Channel assignment to optics is a model choice.
- $r = 1$ creates artificial response pressure. Interpret source+optics selection at $r = 1$ as a consequence of the proxy design, not a growth-leakage detection.

4 Conclusion

The v0.3 SRO audit strengthens the G1DM source–response–optics decomposition. The source floor remains mandatory ($T^D_{\mu\nu} \neq 0$), the response/growth channel is not required ($\mu_{\rm grav} \simeq 1$), and independent S_8 compressed proxies from KiDS-1000 and DES Y3 select an optics/low-redshift lensing-structure component beyond the source floor. This upgrades the G1DM optics branch from a Planck-linked DESI diagnostic to an independently motivated compressed-proxy residual. Production multi-probe confirmation remains pending.